Task 4.5 – Malware Basics

Objective:

Study malware analysis concepts and analyze a benign executable (/usr/bin/ls) inside a Kali Linux virtual machine sandbox using static and dynamic analysis techniques.

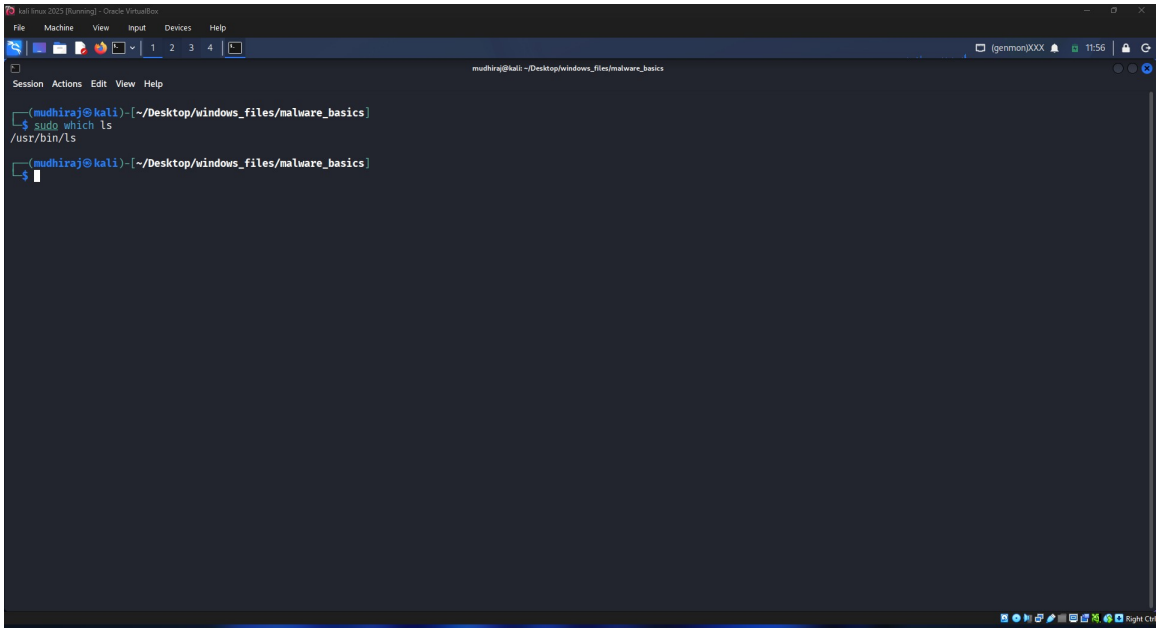
Environment:

• Host OS: Windows
• Guest OS: Kali Linux (VirtualBox)
• Sample: /usr/bin/ls (Benign Linux executable)
• Tools: which, file, strings, readelf, sha256sum, strace, ps

Analysis Steps and Observations:

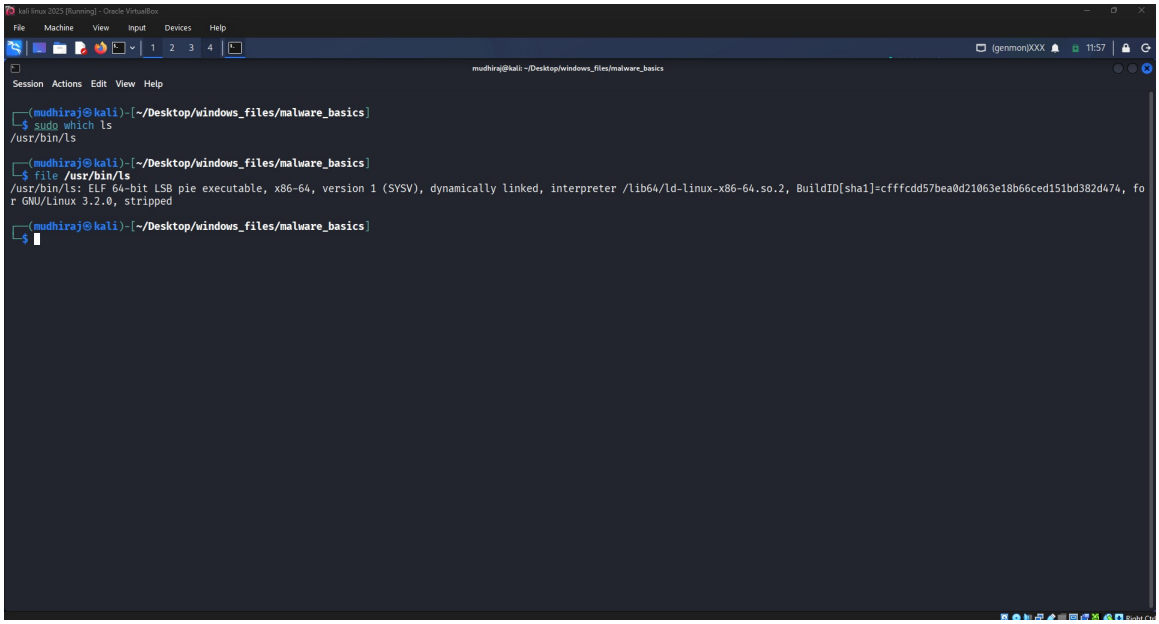
Locate Sample:

The 'which ls' command confirmed the executable location as /usr/bin/ls.



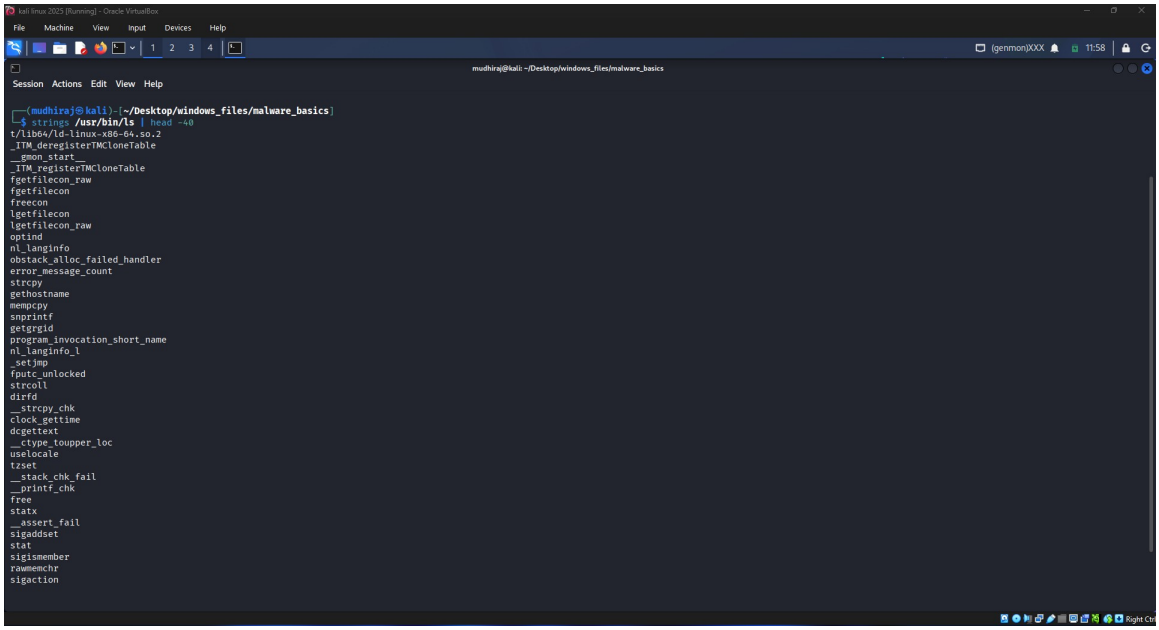
File Identification:

The 'file' command identified it as a 64-bit ELF executable for Linux, dynamically linked and stripped.



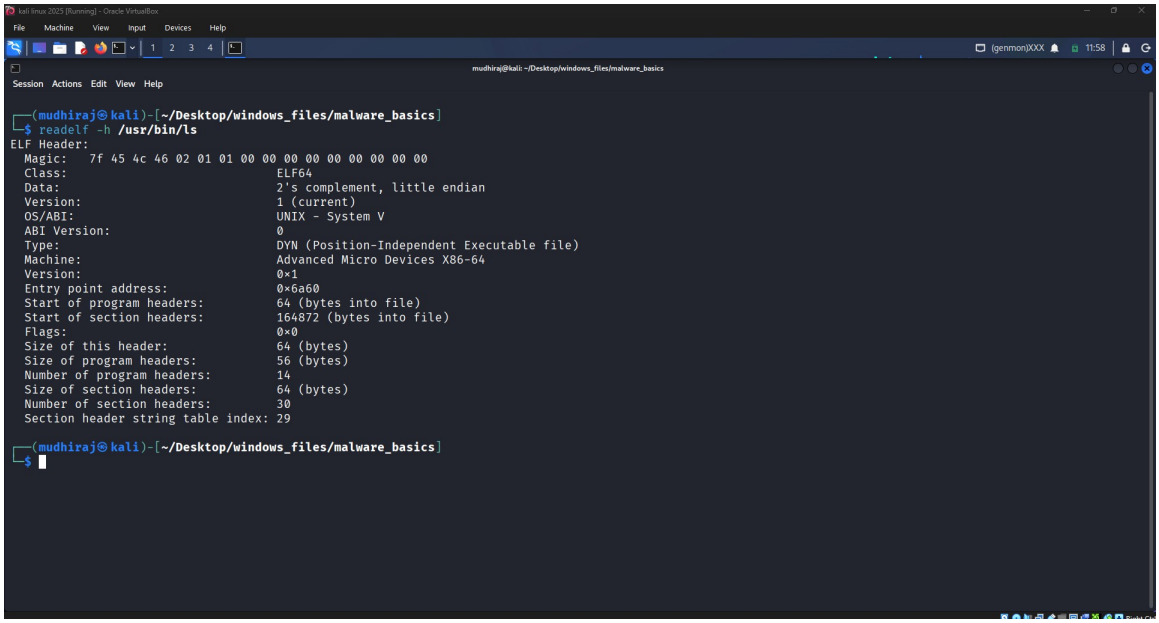
Readable Strings:

The 'strings' command displayed embedded library names and readable symbols used by the executable.



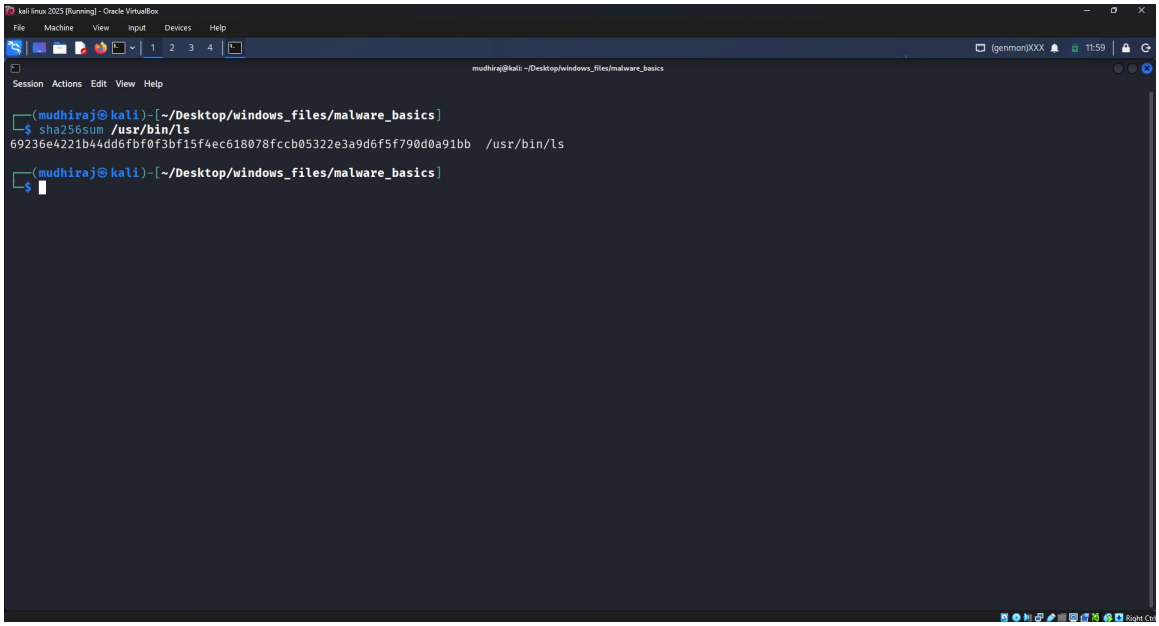
ELF Header:

The 'readelf -h' command displayed the ELF header, architecture (x86-64), entry point, and section information.



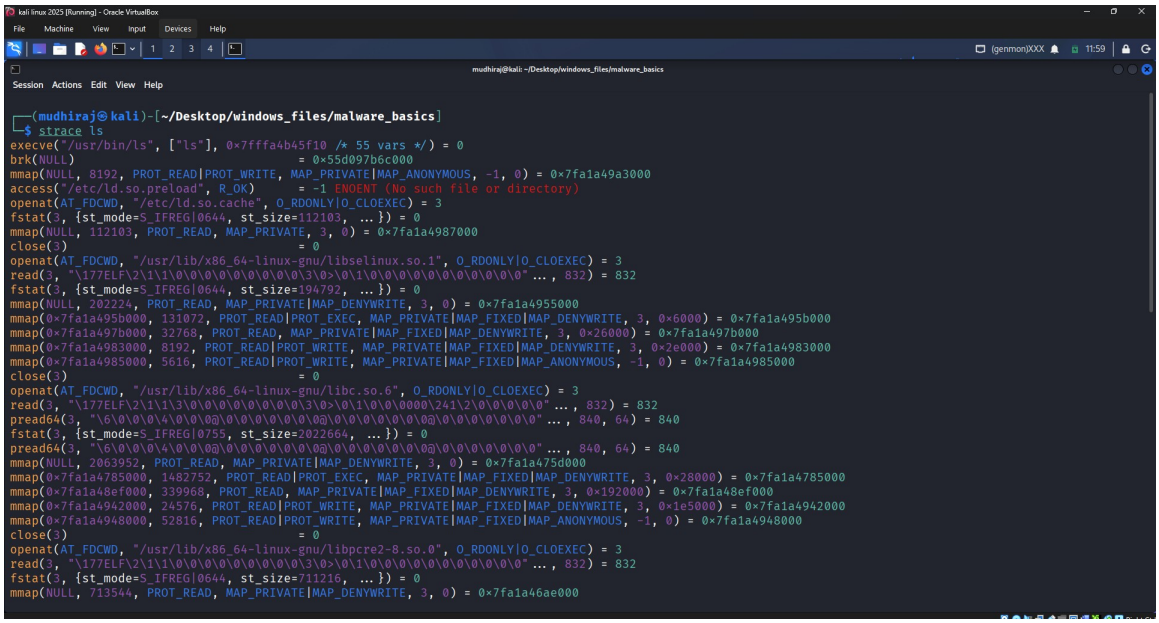
SHA-256 Hash:

A SHA-256 hash was generated to uniquely identify the file and verify integrity.



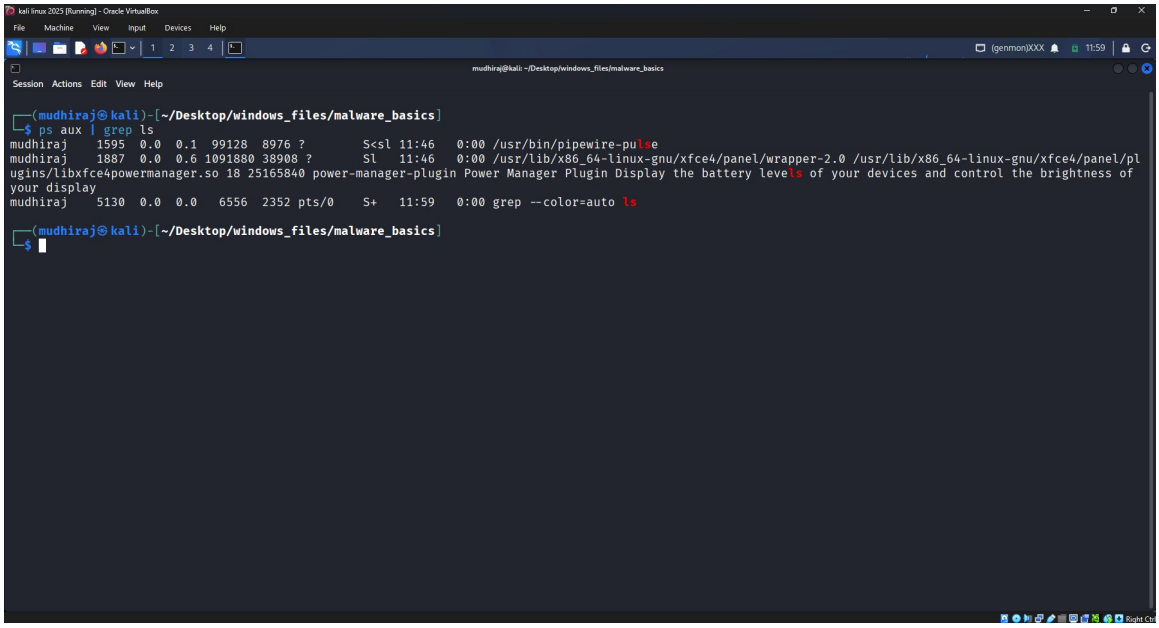
Dynamic Analysis:

The 'strace' command showed system calls such as execve(), openat(), mmap(), read(), close(), indicating normal operating system interaction.



Process Observation:

The 'ps aux | grep ls' command verified process visibility after execution.



Security Assessment:

The analyzed sample is a legitimate Linux system utility and showed no suspicious behavior during analysis.

Observed behavior was limited to loading required libraries, reading configuration files, accessing directories, and interacting with the operating system through expected system calls.

No network communication, persistence mechanisms, privilege escalation, or malicious actions were observed.

Conclusion:

The benign executable /usr/bin/ls was successfully analyzed inside a sandbox environment using both static and dynamic analysis techniques. The executable exhibited expected behavior and no indicators of malicious activity were identified.